

Brandon Ramirez Argueta

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Education

Duke University

August 2022 – May 2026

Double Major in Electrical and Computer Engineering & Computer Science

- **Relevant Coursework:** Data Structures and Algorithms, Computer Architecture, Signals and Systems, Digital Systems, Design and Analysis of Algorithms, Applied Machine Learning, Internet of Things (IoT)

University of Calgary

January 2027 – December 2028

MEng in Electrical and Computer Engineering

Experience

Office of Information Technology, Duke University

May 2024 – July 2024

Full-stack Software Development Intern

- Developed an accessible cost comparison tool for veterans with diabetes on Medicare Part D, sponsored by **Microsoft** and the **VA**, utilizing **Ruby on Rails** for back-end development and **React/Next.js** for front-end development.
- Managed data on insurance, drug pricing, and pharmacy coverage using **PostgreSQL**, integrating information from Medicare Quarterly Pricing files and **VA APIs**.
- Incorporated real-time **API calls** between the front-end, Ruby on Rails back-end, and PostgreSQL database to deliver accurate, up-to-date data and process **OAuth requests**.
- Led daily standups and weekly stakeholder meetings, managing communication and project updates with sponsors.

Maciej A. Mazurowski Lab

June 2023 – January 2024

Undergraduate Research Assistant

- Performed weekly **MRI annotations** across **17 anatomic regions**, contributing to a dataset of 300⁺ **annotated volumes** used to train and fine-tune a **SAM-based foundation model** for automated **bone segmentation**; earned **co-authorship** on the resulting **published research paper**.
- Led an independent project on **image classification** using transfer learning with the **ResNet-18 model**, experimenting with multiple **regularization** and **augmentation** techniques to identify and mitigate **overfitting** in **Convolutional Neural Networks (CNNs)**, improving **model accuracy** and generalization for **medical image analysis** applications.

Projects

Custom 32-bit FPGA Processor & Embedded Claw Machine | [Repository](#)

- Designed and implemented a custom **32-bit pipelined processor** in structural Verilog, including a custom **ALU**, **register file**, **program counter**, **instruction decoder**, and control logic supporting **arithmetic**, **branching**, **memory**, and **jump instructions**. Deployed the processor on a **Xilinx FPGA** operating at 50 MHz to execute assembly programs and control a real-time embedded system.
- Integrated custom processor hardware with FPGA peripherals to interface with joysticks, limit switches, IR sensors, stepper motors, servos, LEDs, and PWM controllers, creating a hardware/software co-designed autonomous claw machine.

Audio Stem Separator | [Repository](#)

- Built an end-to-end **U-Net** based deep learning pipeline in **PyTorch** to separate vocal and instrumental stems from mixed audio, training on **MUSDB18** and a curated dataset with custom **preprocessing** and **augmentation**, and incorporating **Short-Time Fourier Transforms**, **ratio-mask learning**, and **checkpoint training**; evaluated performance using **SI-SDR**, **Mel-L1**, **MAE**, and **MSE** benchmarks.

Duke Motorsports Timing Gates | [Repository](#)

- Designed a **custom PCB** for a **wireless photoelectric timing system**, featuring a transimpedance amplifier and comparator front-end that detects interruptions in a 650nm bandpass-filtered, 100kHz-modulated laser beam. This analog signal chain enables detection response times under 50 to 100 microseconds.
- Iterated the project from a breadboard proof-of-concept (**dual-ESP32**, **Wi-Fi based communication**) to a production-ready **custom PCB**, applying analog front-end design, **RF layout**, and embedded systems development across both hardware revisions.

Smart Pace-Setter Line Following Car | [Repository](#)

- Engineered an **IoT closed-loop line-following RC car** on an **ESP32-S3** running **FreeRTOS**, implementing a weighted-error algorithm for an **8-sensor QTI reflectance array** with **PI speed control** via wheel encoder feedback and PWM servo steering.
- Integrated a **Python Flask** backend and **PostgreSQL** database for real-time **AWS cloud** telemetry logging.

Technical Skills

Languages: Python, Ruby, Java, JavaScript, C, C++, Verilog HDL, MIPS Assembly

Tools & Frameworks: Motec M1 Tune, Altium Designer, AMD Vivado, Git, VS Code, React.js, AWS, Ruby on Rails, Docker, PyCharm, Xcode, PostgreSQL, EAGLE, KiCad, MATLAB, GTKWave, PSpice